Jeremías **Likerman**

Adjunct Researcher, CONICET · Associate Professor, University of Buenos Aires

Institute of Andean Studies ``Don Pablo Groeber" (IDEAN), CONICET--UBA, Argentina

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Education

PhD, 2015 Geological Sciences, University of Buenos Aires (UBA) *Argentina*

Thesis: *Prediction of fracturing on irregular geological surfaces using static methods*. Supervisor: Prof. Ernesto Cristallini.

BSc, 2010 Geological Sciences, University of Buenos Aires (UBA) *Argentina*

Main Career History

May 2026–present Associate Professor, Department of Geological Sciences, UBA *Argentina*

Teaching: Geostatistics and Geoinformatics.

Sep 2025–present Institutional Coordinator, IDEAN, CONICET–UBA *Argentina*

2018–present Adjunct Researcher (Investigador Adjunto), CONICET–IDEAN *Argentina*

Structural geology, geomechanics, geodynamics, natural fractures, and geothermal systems. Research career from 2018; promoted to Investigador Adjunto, November 2023.

May–Sep 2025 Visiting Researcher, GEO3BCN–CSIC *Spain*

Aug–Sep 2024 Visiting Researcher, GEO3BCN–CSIC *Spain*

2015–2017 Postdoctoral Researcher, CONICET and Universitat Politècnica de Catalunya *Argentina/Spain*

2013–2026 Teaching appointments, Department of Geological Sciences, UBA *Argentina*

2013–2014 Visiting Doctoral Researcher, Stanford University *USA*

Major Research Achievements

- More than 15 years of research in structural geology and multiscale numerical modelling, integrating field observations, analogue experiments, geomechanics, and 2D/3D simulations.
- Developed predictive approaches for natural-fracture distribution and connectivity in folded structures and energy reservoirs, including the Yacoraite carbonates and the Vaca Muerta Formation.
- Principal Investigator of the 2026 Spanish supercomputing project on three-dimensional stress transfer between adjacent Andean flat slabs, and Co-PI of a 2026–2029 CONICET project linking tectonic-structural modelling with energy resources.
- Leads and co-leads national and international projects on subduction dynamics, lithospheric deformation, natural fractures, and geothermal systems; coordinator of CONICET–CSIC cooperation on geothermal modelling (2024–2026).
- Supervises early-career researchers and undergraduate, doctoral, and postdoctoral work in structural geology, geomechanics, and numerical modelling.

Selected Recent Major Publications

1. Villarroel, M.A., et al., including **Likerman, J.** (2026). The role of compressional structures in the development of collapse calderas: Insights from analogue models. *Journal of Volcanology and Geothermal Research*, 473, 108582.
2. Gianni, G.M., Gallo, L.C., **Likerman, J.**, et al. (2025). Slab underthrusting is the primary control on flat-slab size. *Science Advances*, 11(28).
3. Correa-Luna, C., Yagupsky, D.L., **Likerman, J.**, & Barcelona, H. (2025). Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play. *Tectonophysics*, 230640.
4. Plotek, B., **Likerman, J.**, & Cristallini, E. (2024). Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model. *Journal of Structural Geology*, 105064.
5. Navarrete, C., et al., including **Likerman, J.** (2024). Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province. *Earth-Science Reviews*, 249, 104651.
6. Gianni, G.M., **Likerman, J.**, Navarrete, C.R., Gianni, C.R., & Zlotnik, S. (2023). Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction. *Nature Communications*, 14, 2083.

Full Curriculum Vitae

Professional Profile

Research profile: Geoscientist specializing in structural geology, geomechanics, and multiscale numerical modelling, with more than 15 years of experience in research, university teaching, fieldwork, and hands-on student training. His research addresses geodynamic, geothermal, and geomechanical processes in Andean systems and energy reservoirs, integrating two- and three-dimensional models to study subduction zones, crustal deformation, volcanic systems with geothermal potential, and natural and induced fractures, with emphasis on the Vaca Muerta Formation.

Teaching, outreach, and knowledge transfer: Teaching experience in Geological Surveying, Geostatistics, Geoinformatics, and Introduction to Geology, including practical work in geological mapping, spatial analysis, geological interpretation, geospatial tools, UAVs, and quantitative methods. Participation in public science communication, professional training, institutional management, and international scientific cooperation.

Academic and Postdoctoral Training

Postdoctoral Fellowship — *Argentina/Spain*
CONICET and UPC — *2015–2017*

- Research topic: thermal evolution of oceanic lithosphere. Supervisors: Dr. Ernesto Cristallini (CONICET–UBA) and Dr. Sergio Zlotnik (UPC).

Visiting Researcher — *USA*
Stanford University — *2013*

- Courses in Structural Geology and MATLAB applied to Geosciences.

PhD in Geological Sciences — *Argentina*
University of Buenos Aires — *2010–2015*

- Thesis: *Prediction of fracturing on irregular geological surfaces using static methods*. Supervisor: Dr. Ernesto Cristallini.

BSc in Geological Sciences — *Argentina*
University of Buenos Aires — *2005–2010*

- Thesis: *Geology and structure of the Tres Cruces compound anticline, Jujuy, Argentina*. Supervisor: Dr. Ernesto Cristallini.

Appointments and Professional Experience

Current appointments

Adjunct Researcher (Investigador Adjunto) — *Argentina*
CONICET–IDEAN — *2018–present*

- Research areas: fracture prediction, geodynamic modelling, and geothermal-potential modelling. CONICET research career from 2018; promoted to Investigador Adjunto, November 2023.

Associate Professor — *Argentina*
Department of Geological Sciences, University of Buenos Aires — *May 2026–present*

- Courses: Geostatistics and Geoinformatics.

Institutional Coordinator — *Argentina*
IDEAN, CONICET–UBA — *Sep 2025–present*

- Council Member since April 2024.

Previous teaching appointment

Teaching Assistant — *Argentina*
Department of Geological Sciences, University of Buenos Aires — *2013–2026*

- Courses: Introduction to Geology, Geological Surveying, and Geostatistics.

Professional courses and field activities

Geology of Unconventional Reservoirs
Argentina

Instructor in charge
2025

- Professional training course focused on the geological characterization of shale reservoirs.

Advanced Research Field Trip
Argentina

Field instructor
2025

- Organizer and instructor for *Perspectives on Vaca Muerta: subsurface attributes analysed from surface exposures*.

Research visits

Visiting Researcher *Spain*
GEO3BCN–CSIC *May–Sep 2025*

- Thermomechanical modelling of volcanic zones for geothermal-potential assessment. Host: Dr Ivone Jiménez-Munt.

Visiting Researcher *Spain*
GEO3BCN–CSIC *Aug–Sep 2024*

- Development of coupled numerical models for volcanic and geothermal systems.

Visiting Researcher *Spain*
BarcelonaTech (UPC) *2015, 2017, 2019, 2022*

- Research stays at the Numerical Computation Laboratory.

Research Projects

2026	Three-dimensional models of stress transfer between adjacent Andean flat slabs (Principal Investigator). AECT-2026-1-0080	*Spain*
2026–2029	Tectonic-structural analysis through analogue and numerical modelling: contributions to energy resources (Co-PI). PIP 11220250100817CO	*Argentina*
2025	Geothermal potential of volcanic zones (Principal Investigator). AECT-2025-1-0009	*Spain*
2025–2026	Structural controls in volcanic systems: magmatic intrusions and damage-zone development associated with hydrocarbon generation (Co-PI). Williams Foundation	*Argentina*
2024–2026	Assessment of geothermal potential in volcanic zones using thermomechanical numerical models (PI). LINCGL2024	*Spain*
2024	Geodynamic modelling of subduction zones and mantle plumes (PI). AECT-2024-2-0027	*Spain*
2023	Geodynamics of the Adria microplate, from mantle to surface (Co-PI). GEO3BCN	*Spain*
2022	Geodynamic modelling of subduction zones (PI). AECT-2022-3-0023	*Spain*
2020	Landscape dynamics and geological hazards associated with dams in mountain regions (Co-PI). PICT-2019-04011	*Argentina*
2020	Analogue and numerical modelling: contributions to tectonic and structural problems (Co-PI). PICT-2019-00997	*Argentina*
2018	Natural fracturing and first-order structures: predictive models applied to Cerro Domuyo (Co-PI). UBACyT	*Argentina*
2017	Geological evolution of the Andes and its economic and environmental impact (Co-PI). PUE 22920160100051	*Argentina*
2016	Numerical modelling of the thermal evolution of oceanic lithosphere (PI). PICT 2015-1226	*Argentina*
2016	Application of analogue and numerical models to tectonic-structural problems (Co-PI). PICT-2016-1407	*Argentina*

Research Supervision

Researcher supervision

Plotek, Berenice *Argentina*
Assistant Researcher *Since Aug. 2025*

- Supervisor: Likerman, J.; co-supervisor: Cristallini, E.

Postdoctoral supervision

Plotek, Berenice *Argentina*
Fault-propagation folds: a mechanical approach using finite-element numerical modelling *2024*

- Supervisors: Cristallini, E. and Likerman, J.

Doctoral supervision

Correea Luna, Clara *Argentina*

Natural fracturing in the Vaca Muerta Formation: characterization and modelling *2023*

- Supervisors: Yagupsky, D. and Likerman, J.

Undergraduate Thesis Supervision

Chandía Sepúlveda, Alexander Ignacio *Argentina*
Tectonic inversion of a metamorphic complex in the Deseado Massif: insights from analogue modelling *In progress*

- Supervisors: Likerman, J. and Navarrete, C.

De Rosa, Tomás *Argentina*
Quaternary deformation in the Fiambalá Valley, Catamarca *In progress*

- Supervisors: Likerman, J. and Sagripanti, L.

Blanco, Lucía *Argentina*
Geothermal-electric potential of the Cerro Blanco Volcanic Complex: influence of rheology and structure through numerical models *In progress*

- Supervisors: Likerman, J. and Lamberti, C.

Celis Santisteban, Celso Angel *Argentina*
Structural evolution of an inverted metamorphic complex in the Deseado Massif: insights from numerical modelling *In progress*

- Supervisors: Likerman, J. and Navarrete, C.

Ruiz, Luciana *Argentina*
Assessment of the geothermal-electric potential of the Cerro Blanco Volcanic Complex using numerical modelling *2026*

- Supervisors: Lamberti, C. and Likerman, J.

Gramajo, Pilar *Argentina*
Quaternary deformation in the Chaschuil Valley, western Sierra de Las Planchadas, Catamarca *2025*

- Supervisors: Likerman, J. and Sagripanti, L.

Acosta, Luana *Argentina*
Geomorphology of the Río Grande valley between Bardas Blancas and Portezuelo del Viento, Mendoza *2023*

- Supervisors: Winocur, D. and Likerman, J.

Mazzini, Martín *Argentina*
Geology of the eastern slope of Cerro Perito Moreno, El Bolsón, Río Negro *2022*

- Supervisors: Likerman, J. and Tobal, J.

Relañez, Gastón *Argentina*
Structure and neotectonics of the Tabladitas area, Jujuy, Argentina *2019*

- Supervisors: Yagupsky, D. and Likerman, J.

Ventisky, Esteban *Argentina*
Geology, structure, and fracturing of the Lajas Formation in the Covunco anticline, Neuquén *2017*

- Supervisors: Likerman, J. and Tobal, J.

Correa Luna, Clara *Argentina*
Structural geology and fracturing in the Tres Cruces Basin, Jujuy *2015*

- Supervisors: Yagupsky, D. and Likerman, J.

Peer-Reviewed Publications

The role of compressional structures in the development of collapse calderas: Insights from analogue models
Villarroel, M. A., Browning, J., Jara, P. P., Harnett, C. E., Holohan, E. P., Marquardt, C. J., Guzman, M., **Likerman, J.**
Journal of Volcanology and Geothermal Research *(2026) p. 108582. Elsevier, 2026*

Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play, Neuquén basin, Argentina
Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H.
Tectonophysics *(2025) p. 230640. Elsevier, 2025*

Slab underthrusting is the primary control on flat slab size
Gianni, G. M., Gallo, L. C., **Likerman, J.**, Echaurren, A., Gianni, C. R., Faccena, C.
Science Advances *11.28 (2025). American Association for the Advancement of Science, 2025*

Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province
Navarrete, C., Gianni, G., Tassara, S., Zaffarana, C., **Likerman, J.**, Márquez, M., Wostbrock, J., Planavsky, N., Tardani, D., Frasette, M. P.
Earth-Science Reviews *249 (2024) p. 104651. Elsevier, 2024*

Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model
Plotek, B., **Likerman, J.**, Cristallini, E.
Journal of Structural Geology *(2024) p. 105064. Elsevier, 2024*

Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction
Gianni, G. M., **Likerman, J.**, Navarrete, C. R., Gianni, C. R., Zlotnik, S.
Nature communications *14.1 (2023) p. 2083. Nature Publishing Group UK London, 2023*

Natural fracture characterization of the Los Catutos Member (Vaca Muerta Formation) in the southern sector of the Sierra de la Vaca Muerta, Neuquén Basin, Argentina
Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H.
Journal of South American Earth Sciences *120 (2022) p. 104081. Elsevier, 2022*

Kinematics of fault-propagation folding: Analysis of velocity fields in numerical modeling simulations
Plotek, B., Heckenbach, E., Brune, S., Cristallini, E., **Likerman, J.**
Journal of Structural Geology *162 (2022) p. 104703. Elsevier, 2022*

The influence of climate on the dynamics of mountain building within the Northern Patagonian Andes
García Morabito, E., Beltrán-Triviño, A., Terrizzano, C. M., Bechis, F., **Likerman, J.**, Von Quadt, A., Ramos, V. A.
Tectonics *40.2 (2021) e2020TC006374. Wiley Online Library, 2021*

The effects of small-scale convection in the shallow lithosphere of the North Atlantic
Likerman, J., Zlotnik, S., Li, C.-F.
Geophysical Journal International *227.3 (2021) pp. 1512–1522. Oxford University Press, 2021*

Fracture network analysis of Yacoraite Formation in the Tres Cruces sub-basin, northwestern Argentina
Luna, C. C., Yagupsky, D. L., **Likerman, J.**
Journal of South American Earth Sciences *(2019) p. 102446. Elsevier, 2019*

Cenozoic intraplate tectonics in Central Patagonia: Record of main Andean phases in a weak upper plate
Gianni, G. M., Echaurren, A., Folguera, A., **Likerman, J.**, Encinas, A., García, H. P. A., Dal Molin, C., Valencia, V. A.
Tectonophysics *721 (2017) pp. 151–166. Elsevier, 2017*

Closure type effects on the structural pattern of an inverted extensional basin of variable width: Results from analogue models
Jara, P., **Likerman, J.**, Charrier, R., Herrera, S., Pinto, L., Villarroel, M., Winocur, D.
Journal of South American Earth Sciences *(2017). Elsevier, 2017*

Climatic and Tectonic forcing on alluvial fans in the Southern Central Andes
Terrizzano, C. M., Morabito, E. G., Christl, M., **Likerman, J.**, Tobal, J., Yamin, M., Zech, R.
Quaternary science reviews *172 (2017) pp. 131–141. Elsevier, 2017*

Structural and statistical characterization of joints and multi-scale faults in an alternating sandstone and shale turbidite sequence at the Santa Susana Field Laboratory: Implications for their effects on groundwater flow and contaminant transport
Cilona, A., Aydin, A., **Likerman, J.**, Parker, B., Cherry, J.
Journal of Structural Geology *85 (2016) pp. 95–114. Elsevier, 2016*

Middle to late Miocene extensional collapse of the North Patagonian Andes (41° 30′–42° S)
Tobal, J. E., Folguera, A., **Likerman, J.**, Naipauer, M., Sellés, D., Boedo, F. L., Ramos, V. A., Gimenez, M.
Tectonophysics *657 (2015) pp. 155–171. Elsevier, 2015*

Geodynamic context for the deposition of coarse-grained deep-water axial channel systems in the Patagonian Andes
Ghiglione, M. C., **Likerman, J.**, Barberón, V., Giambiagi, L. B., Aguirre-Urreta, B., Suarez, F.
Basin Research *26.6 (2014) pp. 726–745. Wiley Online Library, 2014*

Role of basin width variation in tectonic inversion: insight from analogue modelling and implications for the tectonic inversion of the Abanico Basin, 32°–34° S, Central Andes
Jara, P., **Likerman, J.**, Winocur, D., Ghiglione, M. C., Cristallini, E. O., Pinto, L., Charrier, R.

Geological Society, London, Special Publications *399 (2014) SP399–7. Geological Society of London, 2014*

Along-strike structural variations in the Southern Patagonian Andes: Insights from physical modeling
Likerman, J., Burlando, J. F., Cristallini, E. O., Ghiglione, M. C.
Tectonophysics *590 (2013) pp. 106–120. Elsevier, 2013*

Fellowships and Distinctions

2027	**Research Fellowship in Japan** Kobe University	*JSPS*
2025	**SZNet 2025 Chile Field Trip** National Science Foundation	*SZ4D*
2022	**Academic Mobility Fellowship among Partner Institutions** Ibero-American University Association for Postgraduate Studies	*AUIP*
2016	**International Research Stay** National Scientific and Technical Research Council	*CONICET*
2015	**Postdoctoral Fellowship** National Scientific and Technical Research Council	*CONICET*
2013	**Doctoral Research Stay at Stanford University** Fulbright – Bunge & Born Foundation	*Doctoral fellowship*
2013	**Doctoral Fellowship Type II** National Scientific and Technical Research Council	*CONICET*
2010	**Doctoral Fellowship Type I** National Scientific and Technical Research Council	*CONICET*

Skills

Programming and reproducibility: LaTeX, Python, R, MATLAB, HTML5, Git, and reproducible scientific documentation.

Modelling and HPC: Underworld2, LaMEM, GOLEM, ASPECT; analogue modelling, experimental design, PIV, post-processing, MPI, and high-performance-computing workflows.

Spatial analysis: QGIS, GMT, PostGIS; seismic interpretation, geological mapping, and UAV-assisted field methods.

Fieldwork: more than 15 years of experience, extensive knowledge of the Andes, campaign planning and logistics, and team leadership.

Languages: Spanish (native), English (fluent, C2), Italian (A2), Catalan (A2).